



**SARUS AEROSPACE**  
PRIVATE LIMITED · INDIA

USER MANUAL

# Sarus Operation Planner

Ground control station for ArduPilot aircraft. Installation, specifications, and a complete guide to every screen, feature and keyboard shortcut.



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SOFTWARE VERSION  
**1.0.0**

PLATFORM  
**Windows 10 / 11**

DOCUMENT  
**SOP-UM-001**

ISSUED  
**October 2026**

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## 01 Safety and about this manual

### SAFETY FIRST

Take every propeller off before you calibrate, test motors, change parameters on the bench or install firmware. One wrong parameter is enough to crash an aircraft, so save a backup before each session and change one thing at a time. Even during an AUTO mission, keep the radio control transmitter in your hands, ready to take over. Fly only where and how the law allows; in India that means the Drone Rules 2021 and the DGCA's Digital Sky requirements. The software comes without warranty under the GNU GPL v3, and the operator is responsible for every flight.

This manual covers Sarus Operation Planner 1.0.0 for Windows. The program is built on ArduPilot Mission Planner 1.3.83 and keeps every one of its screens, controls and keyboard shortcuts, so Mission Planner tutorials apply here too. What differs is the look, the way parameters are edited (chapter 11), more reliable communication with the aircraft, and where files are kept.

**Conventions:** on-screen names are in **bold**. Menu paths are written **CONFIG → Full Parameter List**. Keys are shown as **Ctrl** + **P**. Parameter names are in MONOSPACE.

## 02 Product overview

Sarus Operation Planner is a ground control station (GCS): Windows software that connects to a drone's flight controller and lets you set up, plan, fly, monitor and analyse ArduPilot aircraft.

### What you can do with it

Before a flight you can install or update the firmware on the flight controller, calibrate the accelerometers, compass, radio and ESCs, set flight modes and failsafes, and tune the aircraft: PIDs, TECS, navigation and the VTOL settings, down to any single parameter. Missions are planned on a map with waypoints, surveys, geofences and rally points.

In the air, the DATA screen gives you a live HUD, map, gauges and quick actions, and the program can speak mode changes, altitude and battery warnings. Afterwards you can replay the telemetry, download the flight controller's own log and analyse both. ArduPilot's simulator lets you fly without hardware, and optional equipment such as GPS, airspeed sensors, rangefinders, gimbals, OSD, CAN devices, ADS-B receivers and telemetry radios is set up from the same program.

### Supported vehicles

| VEHICLE                         | FIRMWARE                | SARUS TEST STATUS IN 1.0.0                           |
|---------------------------------|-------------------------|------------------------------------------------------|
| Fixed wing and QuadPlane (VTOL) | ArduPilot Plane         | Tested in the simulator                              |
| Multicopter and helicopter      | ArduPilot Copter        | Multicopter tested in the simulator                  |
| Rover and boat                  | ArduPilot Rover         | Tested in the simulator                              |
| Submarine, antenna tracker      | ArduSub, AntennaTracker | Supported as in Mission Planner, not tested by Sarus |

Flight controllers: any board that runs ArduPilot, including Pixhawk-class boards (Pixhawk, Cube, Holybro, Matek, CUAV and others).

### What is different from Mission Planner

|             |                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Look        | Sarus Dark (default) and Sarus Light themes, Sarus menu icons and logo. HUD and map colours are unchanged.                                                                                   |
| Parameters  | Out-of-range values give an amber warning instead of a blocking question. Read-only values can be changed after confirmation. Every written value is re-read from the aircraft (chapter 11). |
| Reliability | Replies from the aircraft can no longer be lost when two requests wait at the same time (parameters, commands, missions).                                                                    |
| Coexistence | Installs beside Mission Planner with separate settings, logs, maps and updates.                                                                                                              |
| Privacy     | Anonymous usage statistics are off by default.                                                                                                                                               |
| Airspace    | No Altitude Angel or other airspace service is included, so the program never restricts where you fly. The aircraft's own geofence and failsafes still apply, under your control.            |

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**Network safety** The built-in web server accepts flight commands only from this PC; other devices on the network can still view telemetry.

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## 03 Specifications and requirements

### Software

|                        |                                                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Product                | Sarus Operation Planner                                                                                                                          |
| Version                | 1.0.0 (file version 1.3.83, the Mission Planner base)                                                                                            |
| Publisher              | Sarus Aerospace Pvt Ltd, India · sarusaerospace.in                                                                                               |
| Based on               | ArduPilot Mission Planner 1.3.83                                                                                                                 |
| Communication protocol | MAVLink 1 and 2                                                                                                                                  |
| Connections            | USB/serial (COM ports), TCP, UDP, UDP client, WebSocket                                                                                          |
| Installer              | Windows Installer package (.msi), about 110 MB                                                                                                   |
| Licence                | GNU General Public License v3, source at <a href="https://github.com/LikhitS/SarusOperationPlanner">github.com/LikhitS/SarusOperationPlanner</a> |
| Languages              | English, plus the Mission Planner translations (CONFIG → Planner → UI Language)                                                                  |

### Computer requirements

| ITEM             | MINIMUM                                                                                                                          | RECOMMENDED                         |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Operating system | Windows 10 (64-bit)                                                                                                              | Windows 11 (64-bit)                 |
| .NET Framework   | 4.7.2 (built into Windows 10 version 1803 and later)                                                                             | 4.8                                 |
| Processor        | Dual-core, 2 GHz                                                                                                                 | Quad-core or better                 |
| Memory           | 4 GB                                                                                                                             | 8 GB or more                        |
| Disk             | 500 MB plus map cache and logs                                                                                                   | 5 GB or more free for maps and logs |
| Display          | 1280 × 720                                                                                                                       | 1920 × 1080, 100% scaling           |
| Ports            | One USB port for the flight controller or telemetry radio                                                                        |                                     |
| Internet         | Optional. Needed for firmware downloads, parameter descriptions, online maps and updates. Flying works offline with cached maps. |                                     |

Above 100% Windows display scaling, text looks slightly soft. This is the same in Mission Planner.

## 04 Installation

### 4.1 Download

Download the latest SarusOperationPlanner-x.y.z.msi from the Sarus releases page: [github.com/LikhitS/SarusOperationPlanner/releases](https://github.com/LikhitS/SarusOperationPlanner/releases), or from the Sarus Aerospace website.

### 4.2 Install

- 1 Double-click the downloaded .msi file.
- 2 If Windows shows **Windows protected your PC**, click **More info**, check that the file is the one from Sarus, then click **Run anyway**. This appears until the installer carries a code signature.
- 3 Read and accept the licence (GNU GPL v3), then click **Next**.
- 4 Keep the default folder C:\Program Files (x86)\Sarus Operation Planner or choose another, then click **Install** and approve the administrator prompt.
- 5 When Windows asks to install device drivers (for flight-controller USB ports), click **Install**. These are the same drivers Mission Planner installs.
- 6 Click **Finish**. Start the program from the Start menu: **Sarus Operation Planner**.

#### SILENT INSTALL (IT DEPARTMENTS)

msiexec /i SarusOperationPlanner-1.3.83.msi /qn /norestart installs without questions. In a silent install, Windows adds only the USB drivers whose publisher is already trusted; ArduPilot boards on Windows 10/11 normally use the built-in Windows USB serial driver. Uninstall with msiexec /x and the same file, or from Windows Settings → Apps.

### 4.3 Next to Mission Planner

Sarus Operation Planner and Mission Planner can be installed on the same PC. Each has its own program folder, Start-menu entry, settings, logs, map cache and update channel. Installing or removing one does not affect the other. Only one program can use a given COM port at a time.

### 4.4 Updating

On start, the program checks the Sarus releases on GitHub. When a new version exists, it offers to update. The updater downloads only the files that changed and verifies each file's checksum. You can also check manually in **HELP → Check for Updates**, or install a newer .msi over the current one; settings are kept.

### 4.5 Uninstalling

Windows Settings → Apps → Installed apps → **Sarus Operation Planner** → Uninstall. Your logs, missions and settings in Documents\Sarus Operation Planner are kept; delete that folder by hand if you no longer need them.

## 05 First start and the main window

### 5.1 Questions on first start

| MESSAGE                                           | WHAT TO DO                                                                                                                                                                                                                                       |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows Security: allow network access            | Not shown after a normal installation: the installer already allows the program on every network type, and removes that permission when you uninstall. If you run the portable copy, Windows asks once; tick both boxes and click <b>Allow</b> . |
| Restart to enable a plugin (for example Drone ID) | Restart the program if you want the plugin, or disable it in <b>Ctrl + P</b> if you do not.                                                                                                                                                      |
| Other "show again?" messages                      | Informational. Tick "don't show again" if you prefer.                                                                                                                                                                                            |

### 5.2 The main window

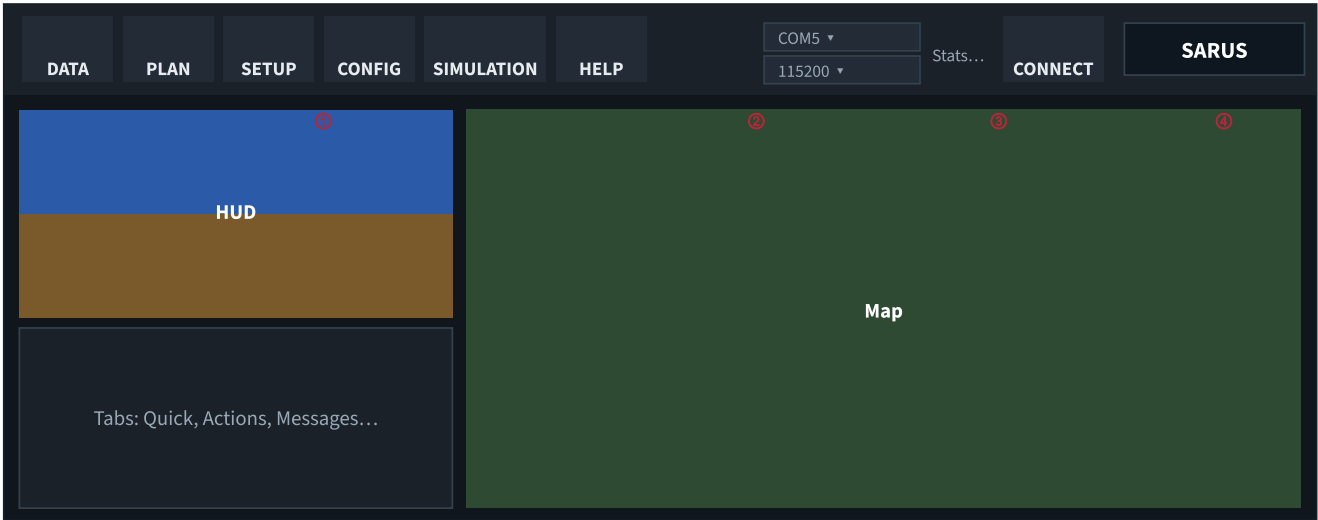


Figure 1. Main window on the DATA screen. ① Screen menu ② Connection type and speed ③ Link statistics and CONNECT ④ Sarus logo (opens sarusaerospace.in).

| MENU       | SCREEN          | USE IT TO                                               | KEY       |
|------------|-----------------|---------------------------------------------------------|-----------|
| DATA       | Flight Data     | Monitor and command the aircraft in flight; replay logs | <b>F2</b> |
| PLAN       | Flight Planner  | Create missions, geofences and rally points             | <b>F3</b> |
| SETUP      | Hardware Config | Install firmware, calibrate, set up hardware            |           |
| CONFIG     | Software Config | Tune, edit parameters, program settings                 | <b>F4</b> |
| SIMULATION | Simulation      | Fly a simulated aircraft (SITL)                         |           |
| HELP       | Help            | Updates, change log, console window                     |           |

The program remembers its window size, theme and last connection.

## 06 Connecting to an aircraft

### 6.1 Connection types

| TYPE (TOP-RIGHT LIST) | USE FOR                                                                      | USUAL SPEED / PORT             |
|-----------------------|------------------------------------------------------------------------------|--------------------------------|
| COMx                  | USB cable to the flight controller, or a USB telemetry radio (SiK, RFD900)   | USB: 115200 · SiK radio: 57600 |
| AUTO                  | Tries the available COM ports and speeds automatically                       | –                              |
| TCP                   | Network link to an aircraft, a companion computer or SITL                    | e.g. 127.0.0.1:5760            |
| UDP                   | Listen for an aircraft broadcasting telemetry (Wi-Fi radios, Herelink, SITL) | 14550                          |
| UDPCl                 | Send to a fixed address and port (UDP client)                                | host and port                  |
| WS                    | WebSocket link                                                               | URL                            |

### 6.2 Connect

- 1 Plug in the flight controller (USB) or the ground telemetry radio, and power the aircraft.
- 2 Choose the COM port (or TCP/UDP) and the speed in the top-right lists.
- 3 Click **CONNECT** or press **F12**.
- 4 The program waits for a heartbeat, then downloads all parameters (a progress window shows this). When it finishes, the HUD moves and the title bar shows the firmware version.

Click **DISCONNECT** (or **F12** again) to disconnect. **Ctrl + T** connects without downloading the parameters (advanced). **Stats...** under the connection lists shows link quality, bytes per second and packet loss.

With several aircraft or components on one link, choose which one to control from the system-ID list that appears in the connection area.

## 07 DATA – Flight Data screen

This is the screen you fly from. The HUD sits at the top left, the tabs below it and the map on the right; drag the dividers to change the proportions.

### 7.1 HUD (head-up display)

The HUD draws the artificial horizon with roll and pitch, plus heading, airspeed and ground speed (left tape), altitude (right tape), vertical speed, flight mode, armed state, GPS status, battery, link quality, distance to waypoint and current waypoint, plus warnings such as PreArm messages, EKF problems and low battery. HUD colours are those of Mission Planner.

**Right-click the HUD** for: **Video** (start a camera, MJPEG or GStreamer source, Herelink video, record HUD to AVI), **Set Aspect Ratio**, **User Items** (add any telemetry value to the HUD), **Swap With Map**, **Ground Color**, **Battery Cell Voltage**, **Show icons** and **Russian Hud**.

### 7.2 Tabs

Press **Ctrl** + **1** to **Ctrl** + **0** to jump to the 1st to 10th visible tab. Right-click a tab strip to choose which tabs are shown.

| TAB                     | CONTENTS                                                                                                                                                                                                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quick</b>            | Large-number display of selected values (altitude, speed, distance, yaw, vertical speed...). Double-click a value to choose another. Right-click to set the number of values, or undock.                                                      |
| <b>Actions</b>          | Do Action (any command) · Set WP · Set Mode · Set Mount · Arm/Disarm · Auto · Loiter · RTL · Restart Mission · Resume Mission · Abort Landing · Set Home Alt · Clear Track · Raw Sensor View · Joystick · Message · Change Speed · Change Alt |
| <b>Messages</b>         | All text messages from the aircraft, with time stamps (PreArm, EKF, mode changes, errors)                                                                                                                                                     |
| <b>Actions (simple)</b> | Reduced set of large action buttons                                                                                                                                                                                                           |
| <b>PreFlight</b>        | Configurable pre-flight checklist with live checks                                                                                                                                                                                            |
| <b>Gauges</b>           | Analogue speed, altitude, vertical-speed and heading gauges                                                                                                                                                                                   |
| <b>Transponder</b>      | ADS-B transponder control: squawk, flight ID, IDENT, mode (STBY / ON / ALT)                                                                                                                                                                   |
| <b>Status</b>           | Every telemetry value the program receives, in one list                                                                                                                                                                                       |
| <b>Servo/Relay</b>      | Toggle relays and drive servo outputs (low / high / toggle)                                                                                                                                                                                   |
| <b>Aux Function</b>     | Trigger ArduPilot auxiliary functions (as an RC switch would)                                                                                                                                                                                 |
| <b>Scripts</b>          | Run Python scripts on the ground station                                                                                                                                                                                                      |
| <b>Payload Control</b>  | Gimbal roll, pitch and pan, video control, reset position                                                                                                                                                                                     |
| <b>Telemetry Logs</b>   | Load and replay a .tlog (Play/Pause, speed 0.1× to 10×, slider), convert tlog to KML or graph                                                                                                                                                 |

| TAB                   | CONTENTS                                                                                                                                                |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DataFlash Logs</b> | Download DataFlash Log Via Mavlink · Review a Log · Auto Analysis · Create KML + gpx · Convert .Bin to .Log · Create Matlab File · Geo Reference Images |

## 7.3 Map

On the map you see the aircraft and its track, home, the mission, geofence, rally points, ADS-B traffic and the planned path. Scroll to zoom; drag to pan. Hold **Ctrl** and click the map to send the aircraft there (same as **Fly To Here**, guided mode).

**Right-click the map:** Fly To Here · Fly To Here Alt · Fly To Coords · TakeOff · Add Poi · Point Camera Here / Point Camera Coords · Trigger Camera NOW · Set Home Here · Set EKF Origin Here · Flight Planner · Camera Overlap · Jump To Tag. Checkboxes under the map control **Auto Pan**, **Tuning** (live graph of selected values) and the zoom level.

### GUIDED COMMANDS

Fly To Here changes the aircraft to GUIDED mode. Check the altitude before you send it.

## 7.4 Log playback keys

While a telemetry log is loaded: **Space** plays or pauses, numeric-keypad **+** speeds up and **-** slows down.

# 08 PLAN – Flight Planner screen

Missions, geofences and rally points are created, loaded and sent from here. The map takes most of the screen, with the mission table along the bottom and the action panel on the right.

## 8.1 Making a mission

- 1 Check **Home Location** (filled in from the aircraft when connected).
- 2 Set the defaults: **WP Radius**, **Loiter Radius**, **Default Alt**, and the altitude frame (**Relative**, **Absolute** or **Terrain**). Tick **Verify Height** to use terrain data.
- 3 Left-click the map to add waypoints. Drag a waypoint to move it.
- 4 In the table, change each row's **Command** (WAYPOINT, VTOL\_TAKEOFF, VTOL\_LAND, LOITER\_TURNS, DO\_JUMP, DO\_SET\_SERVO, DO\_CHANGE\_SPEED...) and its parameters, latitude, longitude and altitude. The column headers change to explain the parameters of the selected command. Use the arrows to reorder and **X** to delete.
- 5 Click **Write** (**Ctrl** + **Shift** + **W**) to send the mission to the aircraft, then **Read** (**Ctrl** + **Shift** + **R**) to confirm it. **Write Fast** (**Ctrl** + **Shift** + **F**) sends without waiting for each item.
- 6 Save the mission with **Ctrl** + **S**; open one with **Ctrl** + **O**. **Ctrl** + **Z** undoes the last change.

The status bar shows total distance, distance between points, and an **Alt Warn** indication. Use **Elevation Graph** (right-click) to check terrain clearance.

## 8.2 Right-click menu on the map

|                                  |                                                                                                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Insert / Delete                  | Insert Wp · Insert Spline WP · Delete WP · Add Below                                                                                                                                                   |
| Loiter, Jump, RTL, Land, Takeoff | Add a loiter (Forever, Time, Circles), a jump (Start, WP #), RTL, Land or Takeoff command at the clicked point                                                                                         |
| DO_SET_ROI                       | Point the camera or vehicle at a location                                                                                                                                                              |
| Clear Mission                    | Remove all waypoints (on screen only until you Write)                                                                                                                                                  |
| Draw a Polygon                   | Draw, clear, save or load a polygon; from SHP or from current waypoints; offset; show its area                                                                                                         |
| Geo-Fence                        | Upload, download, set return location, load from or save to file, clear; inclusion and exclusion fences                                                                                                |
| Rally Points                     | Set rally point, upload, download, save to or load from file, clear                                                                                                                                    |
| Auto WP                          | Survey (Grid) · Create Circle Survey · Create Wp Circle · Create Spline Circle · Text                                                                                                                  |
| Map Tool                         | Measure Distance · Rotate Map · Zoom To · Prefetch (download map tiles for offline use) · Prefetch WP Path · KML Overlay · Elevation Graph · Reverse WPs · Modify Alt · Enter UTM Coord · GDAL opacity |
| File Load/Save                   | Load WP File · Load and Append · Save WP File · Load KML File · Load SHP File                                                                                                                          |
| POI                              | Add, delete, edit points of interest                                                                                                                                                                   |
| Other                            | Set Home Here · Tracker Home · Switch Docking · Zoom to Vehicle / Mission / Home                                                                                                                       |

**Selecting several waypoints:** hold **Ctrl** and drag a box, or **Ctrl** +click markers; then drag to move them together.

**Geofence and rally editing:** use the drop-down on the right of the screen (**MISSION** / **FENCE** / **RALLY**) to edit those lists with the same table and map tools. Fence points can be inclusion or exclusion zones (polygons or circles).

**Map type:** choose the map provider from the list on the right. **Ctrl** + **X** opens the map cache manager.

## 09 SETUP – Hardware setup

The pages are listed on the left. Those that need a connected aircraft stay greyed out until you connect.

### 9.1 Install Firmware

- 1 **Disconnect** (do not press CONNECT). Plug the flight controller into USB.
- 2 Open **SETUP** → **Install Firmware**. The page shows the latest release for each vehicle (Plane, Copter, Heli, Rover, Sub, Antenna Tracker). It reads ArduPilot's live firmware list together with the Sarus firmware list, so new releases of either appear the next time the program starts.
- 3 Click the vehicle (for a QuadPlane: **Plane**). For Copter, choose the frame type when asked. If both an ArduPilot and a Sarus build exist for your board, a selection window lists them by file address; the Sarus files are the ones whose names start with Sarus-.
- 4 Follow the prompts (unplug and replug the board if asked). Wait for "Upload Done". Never unplug during flashing.

**All Options** lists every version and board. **Beta firmwares** offers test releases. **Load custom firmware** flashes a .apj file you built or were given. **Bootloader Update** / **Force Bootloader** update the board's bootloader (advanced). **Install Firmware Legacy** is the older version of this page.

### 9.2 Mandatory Hardware

|                               |                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frame Type</b>             | Copter frame class and type (Quad X, Hexa...). For QuadPlane, set Q_FRAME_CLASS / Q_FRAME_TYPE in the parameter list.                             |
| <b>Accel Calibration</b>      | Calibrate Accel (six positions: level, left, right, nose down, nose up, back), Calibrate Level, Simple Accel Cal                                  |
| <b>Compass</b>                | Priority order, enable/disable compasses, orientation, Onboard Mag Calibration (rotate the aircraft on all axes until done), Large Vehicle MagCal |
| <b>Radio Calibration</b>      | Move all sticks and switches to their ends; Click when Done. Check each channel moves the right way.                                              |
| <b>Servo Output</b>           | Function, min, trim, max and reverse for each output                                                                                              |
| <b>Serial Ports</b>           | Protocol and baud rate for each serial port                                                                                                       |
| <b>ESC Calibration</b>        | ESC calibration and motor PWM type (Copter)                                                                                                       |
| <b>Flight Modes</b>           | Six modes for the mode switch; current mode and PWM shown live; Save Modes                                                                        |
| <b>FailSafe</b>               | Radio, battery (low and critical voltage / mAh) and GCS failsafes and their actions; live RC input and output display                             |
| <b>Initial Tune Parameter</b> | Starting values from propeller size, battery cells and voltages (Copter)                                                                          |
| <b>HW ID</b>                  | Hardware IDs of the connected sensors                                                                                                             |
| <b>ADSB</b>                   | ADSB receiver setup                                                                                                                               |

## 9.3 Optional Hardware

RTK/GPS Inject (RTCM corrections) · SiK Radio (configure SiK telemetry radios) · CubeID Update · CAN GPS Order · Battery Monitor and Battery Monitor 2 · DroneCAN/UAVCAN · Joystick · Compass/Motor Calib · Range Finder · Airspeed · PX4Flow · Optical Flow · OSD · Camera Gimbal · Antenna tracker · Motor Test · Bluetooth Setup · Parachute · ESP8266 Setup · FFT Setup.

### MOTOR TEST

Props off. Set **Throttle %** (start at 5%) and **Duration**, then test each motor. Check the order and direction against your frame diagram. Set **Motor Spin Arm** and **Set Motor Spin Min** set the spin-when-armed and minimum-flying outputs.

## 9.4 Advanced

**Terminal** (text console to the board or a companion computer) and **Script REPL** (interactive Lua on the flight controller).

# 10 CONFIG – Software configuration and tuning

| PAGE                                     | CONTENTS                                                                                                                                                                                                  |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Flight Modes</b>                      | Same as SETUP → Flight Modes                                                                                                                                                                              |
| <b>GeoFence</b>                          | Fence enable, type, action, maximum altitude, radius                                                                                                                                                      |
| <b>Basic Tuning</b>                      | Simple sliders for responsiveness (Copter); key rates and limits (Plane)                                                                                                                                  |
| <b>Extended Tuning</b>                   | Copter: rate and angle PIDs, navigation. Plane: servo roll/pitch/yaw PIDs, L1 turn control, <b>TECS</b> (climb max, sink min, sink max, time constant, damping), airspeed, throttle and navigation angles |
| <b>QP Extended Tuning</b>                | QuadPlane VTOL PIDs and navigation (Q_ parameters), shown for Plane with QuadPlane enabled                                                                                                                |
| <b>Standard Params / Advanced Params</b> | Parameters with descriptions, units, ranges and drop-down choices                                                                                                                                         |
| <b>Onboard OSD</b>                       | Arrange items on the on-screen display                                                                                                                                                                    |
| <b>MAVftp</b>                            | Browse files on the flight controller's SD card: download, upload, delete (logs, Lua scripts, terrain)                                                                                                    |
| <b>User Params</b>                       | Your own short list of parameters to edit                                                                                                                                                                 |
| <b>Full Parameter List</b>               | Every parameter in a searchable table (chapter 11)                                                                                                                                                        |
| <b>Planner</b>                           | Program settings (below)                                                                                                                                                                                  |

10.1 Planner settings

|                 |                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display         | Theme (Sarus Dark, Sarus Light, Mission Planner themes, Custom) · Layout (Basic / Advanced) · UI Language · HUD options · Speed, distance and altitude units                                   |
| Speech          | Enable Speech; spoken mode, waypoint, battery warning, altitude, arm/disarm, 30-second interval message; Only when Armed                                                                       |
| Map             | Map Follow, map rotated to follow the plane, track length, aircraft icon, COG, heading, nav bearing, turn radius, target, ADS-B, airports, TFRs, no-fly zones, map access mode, open map cache |
| Telemetry Rates | Requested rates for attitude, position, mode/status, RC and sensors (lower on slow radios)                                                                                                     |
| Connection      | Reset on USB connect (DTR), GCS ID, params download in background, load waypoints on connect, slow-computer mode                                                                               |
| Other           | Log path · Video device · Joystick setup · Beta Updates · OptOut Anon Stats (ticked by default in Sarus) · Password Protect Config · Testing Screen · Mavlink Message Debug                    |

## 11 Parameter editing in Sarus

Open **CONFIG** → **Full Parameter List**. Every parameter is shown with its value, units, recommended range, and description.

### 11.1 Buttons

|                                      |                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Write Params</b>                  | Send all changed values (highlighted) to the aircraft                                                |
| <b>Refresh Params</b>                | Download all values from the aircraft again ( <b>F5</b> anywhere)                                    |
| <b>Save to file / Load from file</b> | Save all values to a .param file; load a file (loaded values are marked as changed until you Write)  |
| <b>Compare Params</b>                | Compare the aircraft with a file and choose which values to take                                     |
| <b>Load Presaved</b>                 | Load a ready-made parameter set for a known frame                                                    |
| <b>Reset to Default</b>              | Reset all parameters to firmware defaults (the aircraft must be recalibrated)                        |
| <b>Commit Params</b>                 | Hidden by default. Tells the aircraft to store written values permanently (same as <b>Ctrl + Y</b> ) |
| <b>Search · Modified</b>             | Filter by name or description · show only values you have changed and not yet written                |

### 11.2 Sarus behaviour

| SITUATION                                  | WHAT SARUS DOES                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Value outside the recommended range</b> | The value is accepted. The cell turns <b>amber</b> and its tooltip shows the recommended range. Nothing is blocked; you decide.                                                                                              |
| <b>Read-only parameter</b>                 | A warning asks you to confirm. If you choose Yes, the value is sent.                                                                                                                                                         |
| <b>After Write Params</b>                  | Sarus re-reads every written value from the aircraft. If the aircraft limited, rounded or rejected a value, the window <b>Values changed by aircraft</b> lists it, and the table shows the value the aircraft actually uses. |

#### KNOWN FIRMWARE BEHAVIOURS

ArduPilot does not always keep what it acknowledges. While the loiter controller is running, in QLOITER for example, it caps Q\_LOIT\_ACC\_MAX\_M and LOIT\_ACC\_MAX\_M at what the maximum lean angle allows; 9.81 then becomes about 5.66. EK3\_BCN\_DELAY, EK3\_FLOW\_DELAY and LAND\_THR\_SLEW are stored in 8 bits and stop at 127. The firmware overwrites BARO\*\_GND\_PRESS on its own, and MIS\_TOTAL cannot be written at all. Some settings, Q\_ENABLE, the frame class and the serial protocols among them, only take effect after the flight controller has been rebooted and the parameters refreshed with **F5**. In each of these cases Sarus shows you the value the aircraft actually holds.

**GOOD PRACTICE**

Save to file before you change anything. Change a few values at a time. After a tuning session, save again with a descriptive file name and date.

## 12 SIMULATION

The simulator runs ArduPilot's real firmware (SITL) on your PC, so you can fly without any hardware. We use it for training, for rehearsing missions and for trying out parameter changes before they go near an aircraft.

- 1 Drag the **Home Location** marker to where you want to fly, and set the heading.
- 2 Choose a vehicle: **Plane**, **Multirotor**, **Helicopter** or **Rover**. For a QuadPlane, open the advanced options and choose a quadplane model.
- 3 The firmware is downloaded and started; the program connects automatically.

Advanced options: **Model**, **Sim Speed**, **Wipe** (reset parameters), **Extra command line**, and swarm modes (several simulated aircraft).

## 13 HELP and updates

**Check for Updates** · **Check for BETA Updates** · **Change Log** (opens the Sarus release notes) · **Show Console Window (restart)** (shows diagnostic output, useful for support).

## 14 Keyboard shortcuts

Every shortcut works exactly as in Mission Planner. The list below was taken from the Sarus source code, and the automated tests press each one.

### Anywhere in the program

| KEYS            | ACTION                                                                 |
|-----------------|------------------------------------------------------------------------|
| <b>F2</b>       | DATA (Flight Data) screen                                              |
| <b>F3</b>       | PLAN (Flight Planner) screen                                           |
| <b>F4</b>       | CONFIG (Software Config) screen                                        |
| <b>F5</b>       | Refresh all parameters from the aircraft and reload the current screen |
| <b>F12</b>      | Connect / disconnect                                                   |
| <b>Ctrl + F</b> | Advanced tools window (special functions and test tools)               |
| <b>Ctrl + P</b> | Plugins: enable or disable plugins                                     |
| <b>Ctrl + G</b> | NMEA output: send the aircraft position as GPS NMEA to a serial port   |
| <b>Ctrl + X</b> | Map cache manager                                                      |
| <b>Ctrl + L</b> | Spectrogram (vibration / FFT view)                                     |
| <b>Ctrl + W</b> | Radio propagation (coverage) settings                                  |
| <b>Ctrl + Z</b> | Camera test (outside PLAN; on PLAN it is Undo)                         |
| <b>Ctrl + T</b> | Connect without downloading parameters                                 |
| <b>Ctrl + Y</b> | Tell the aircraft to save its parameters to permanent storage          |
| <b>Ctrl + J</b> | DevOps window (developer diagnostics)                                  |

DATA screen

|                                     |                           |
|-------------------------------------|---------------------------|
| <b>Ctrl</b> + <b>1</b> ... <b>0</b> | Select tab 1 to 10        |
| <b>Space</b>                        | Play / pause log playback |
| <b>Keypad</b> <b>+</b> / <b>-</b>   | Faster / slower playback  |
| <b>Ctrl</b> +click map              | Fly To Here               |

PLAN screen

|                                       |                                                             |
|---------------------------------------|-------------------------------------------------------------|
| <b>Ctrl</b> + <b>Z</b>                | Undo                                                        |
| <b>Ctrl</b> + <b>O</b> / <b>S</b>     | Open / save mission file                                    |
| <b>Ctrl</b> + <b>Shift</b> + <b>W</b> | Write mission                                               |
| <b>Ctrl</b> + <b>Shift</b> + <b>F</b> | Write Fast                                                  |
| <b>Ctrl</b> + <b>Shift</b> + <b>R</b> | Read mission                                                |
| <b>Ctrl</b> + <b>N</b>                | Toggle map drawing mode (GDI)                               |
| <b>Ctrl</b> + <b>M</b>                | Copy the plan into the program's mission memory (no upload) |
| <b>Ctrl</b> +drag / click             | Select several waypoints                                    |

## 15 QuadPlane (VTOL) quick-start workflow

A suggested order for setting up a new QuadPlane. Props off until step 10.

- 1 **Firmware:** SETUP → Install Firmware → Plane.
- 2 **Connect** over USB (115200) and let parameters download.
- 3 **Enable QuadPlane:** set Q\_ENABLE = 1 in the Full Parameter List, Write, reboot the board, press **F5**. Set Q\_FRAME\_CLASS and Q\_FRAME\_TYPE, Write, reboot.
- 4 **Outputs:** SETUP → Servo Output: assign motors 1–4, aileron, elevator, rudder and throttle functions; check reverse settings.
- 5 **Calibrate:** Accel Calibration, Compass, Radio Calibration. Calibrate the airspeed sensor (Optional Hardware → Airspeed).
- 6 **Flight modes:** include QHOVER, QLOITER, FBWA and a return mode (RTL/QRTL).
- 7 **Failsafes:** radio, battery and GCS failsafes, with VTOL-appropriate actions.
- 8 **Motor Test** at 5%: order and direction for each VTOL motor; check surfaces move correctly in FBWA by hand.
- 9 **Save a parameter file** as your baseline.
- 10 **First flight:** follow the Sarus Field Test Checklist (QHOVER → QLOITER → transition → TECS check → small AUTO mission with VTOL\_TAKEOFF and VTOL\_LAND).
- 11 **Tune** with CONFIG → Extended Tuning (TECS), QP Extended Tuning, and the Full Parameter List; review logs after each flight.

### Key QuadPlane and TECS parameters

| PARAMETER                     | MEANING                                                   |
|-------------------------------|-----------------------------------------------------------|
| Q_ENABLE                      | Turn QuadPlane support on                                 |
| Q_FRAME_CLASS / Q_FRAME_TYPE  | VTOL motor layout                                         |
| Q_ASSIST_SPEED                | Below this airspeed, VTOL motors help in fixed-wing modes |
| ARSPD_FBW_MIN / ARSPD_FBW_MAX | Minimum and maximum airspeed in fixed-wing flight         |
| Q_TRANSITION_MS               | Time to transition from VTOL to forward flight            |
| TECS_CLMB_MAX                 | Maximum climb rate (m/s)                                  |
| TECS_SINK_MIN / TECS_SINK_MAX | Minimum and maximum sink rate (m/s)                       |
| TECS_TIME_CONST               | How quickly TECS corrects height and speed errors         |
| TECS_SPDWEIGHT                | Priority between speed and height                         |
| Q_RTL_MODE, Q_OPTIONS         | VTOL return and landing behaviour, options                |

Parameter names change between ArduPilot versions. Always check the description shown in the Full Parameter List for your firmware.

## 16 Logs and files

| WHAT                             | WHERE                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------|
| Telemetry logs (.tlog, .rlog)    | Documents\Sarus Operation Planner\logs\<vehicle>\<id>\, one per connection, recorded automatically |
| Downloaded DataFlash logs (.bin) | Same logs folder                                                                                   |
| Program settings                 | Documents\Sarus Operation Planner\config.xml                                                       |
| Map cache, terrain (SRTM)        | Documents\Sarus Operation Planner\gmapcache, \srtm                                                 |
| Program diagnostic log           | C:\ProgramData\Sarus Operation Planner\SarusOperationPlanner.log                                   |
| Program files                    | C:\Program Files (x86)\Sarus Operation Planner\                                                    |

The log folder can be changed in CONFIG → Planner → Log Path. The installer registers .tlog, .bin and .log files, so you can open them with Sarus Operation Planner from Windows Explorer (right-click → Open with).

### Telemetry logs and DataFlash logs

The **telemetry log** (.tlog) holds what the ground station received over the radio and can be replayed on the DATA screen. The flight controller writes its own **DataFlash log** (.bin), at full rate and with far more detail; download it after flight (DATA → DataFlash Logs) and open it with **Review a Log** to plot values, or use **Auto Analysis**.

## 17 Troubleshooting

| PROBLEM                             | WHAT TO CHECK                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| No COM port in the list             | Cable is a data cable; drivers installed (reinstall Sarus); try another USB port; check Windows Device Manager.                            |
| "No heartbeat" / connection timeout | Correct port and speed (USB 115200, SiK 57600); the port is not used by Mission Planner or another program; radios are paired and powered. |
| Parameter download stops            | Weak radio link: move closer or use USB, or reduce telemetry rates (CONFIG → Planner). Press <b>F5</b> to retry.                           |
| Cannot arm                          | Read the PreArm message on the HUD or in the Messages tab and fix its cause; never disable arming checks to fly.                           |
| "Values changed by aircraft"        | The firmware limited or rejected a value. The table shows the real value. See chapter 11.2.                                                |

| PROBLEM                     | WHAT TO CHECK                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Firmware page empty         | Internet connection needed; disconnect from the aircraft first.                                                                    |
| Map is grey                 | No internet and no cached tiles for that area. Prefetch maps before going to the field (PLAN → right-click → Map Tool → Prefetch). |
| "Windows protected your PC" | See 4.2. Appears until the installer is code-signed.                                                                               |
| Blurry text                 | Windows display scaling above 100%. Same as Mission Planner.                                                                       |
| Program closes or freezes   | Send the diagnostic log (chapter 16) and the telemetry log to Sarus support.                                                       |

## 18 Privacy, licence and support

### Network connections the program makes

|                         |                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------|
| ArduPilot servers       | Firmware lists and downloads, parameter descriptions, SITL binaries                    |
| GitHub (Sarus releases) | Update checks and update downloads                                                     |
| Map providers           | Map tiles and elevation data for the area you view (same providers as Mission Planner) |
| Optional plugins        | Only if you enable them (for example internet ADS-B traffic)                           |
| Usage statistics        | Off by default. Nothing is sent unless you turn it on.                                 |

Telemetry, logs, missions and parameters stay on your computer unless you share them.

### Licence

Sarus Operation Planner is free software under the GNU General Public License version 3. You may copy, change and redistribute it under the terms of that licence. It is provided **without any warranty**. Complete source code: [github.com/Likhits/SarusOperationPlanner](https://github.com/Likhits/SarusOperationPlanner).

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### Support

Sarus Aerospace Pvt Ltd · [www.sarusaerospace.in](http://www.sarusaerospace.in). When reporting a problem, include the Sarus version (HELP screen), firmware version, the steps to reproduce it, the diagnostic log and the telemetry log.

## 19 Abbreviations

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| ADS-B                           | Automatic Dependent Surveillance–Broadcast (aircraft position broadcasts) |
| AHRS                            | Attitude and Heading Reference System                                     |
| CAN                             | Controller Area Network (DroneCAN / UAVCAN peripherals)                   |
| COG                             | Course Over Ground                                                        |
| DGCA                            | Directorate General of Civil Aviation (India)                             |
| EKF                             | Extended Kalman Filter (position and attitude estimator)                  |
| ESC                             | Electronic Speed Controller                                               |
| FBWA / FBWB                     | Fly-By-Wire A / B (assisted fixed-wing modes)                             |
| FFT                             | Fast Fourier Transform (vibration analysis, harmonic notch)               |
| GCS                             | Ground Control Station                                                    |
| GPL                             | GNU General Public License                                                |
| HDOP                            | Horizontal Dilution of Precision (GPS accuracy, lower is better)          |
| HUD                             | Head-Up Display                                                           |
| KML / SHP                       | Google Earth file / shapefile (map shapes)                                |
| MAVLink                         | Micro Air Vehicle Link (aircraft–ground protocol)                         |
| MSI                             | Microsoft Installer package                                               |
| NMEA                            | GPS sentence format (National Marine Electronics Association)             |
| OSD                             | On-Screen Display (video overlay)                                         |
| PID                             | Proportional–Integral–Derivative control loop                             |
| POI / ROI                       | Point / Region Of Interest                                                |
| PWM                             | Pulse Width Modulation (servo and RC signals, in microseconds)            |
| QHOVER / QLOITER / QLAND / QRTL | QuadPlane VTOL modes: hover, hold position, land, return to launch        |
| RC                              | Radio Control                                                             |
| REPL                            | Read–Eval–Print Loop (interactive script console)                         |
| RTCM / RTK                      | GPS correction format / Real-Time Kinematic GPS                           |
| RTL                             | Return To Launch                                                          |
| SiK                             | Common open-source telemetry radio firmware                               |
| SITL                            | Software In The Loop (simulator)                                          |
| SRTM                            | Shuttle Radar Topography Mission (terrain elevation data)                 |
| TECS                            | Total Energy Control System (ArduPlane speed and height control)          |
| tlog / bin                      | Telemetry log / DataFlash (on-board) log                                  |
| UTM                             | Universal Transverse Mercator (grid coordinates)                          |
| VTOL                            | Vertical Take-Off and Landing                                             |
| WP                              | Waypoint                                                                  |